E-SNPs&GO: Embedding of protein sequence and function improves the prediction of human pathogenic variants

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Background

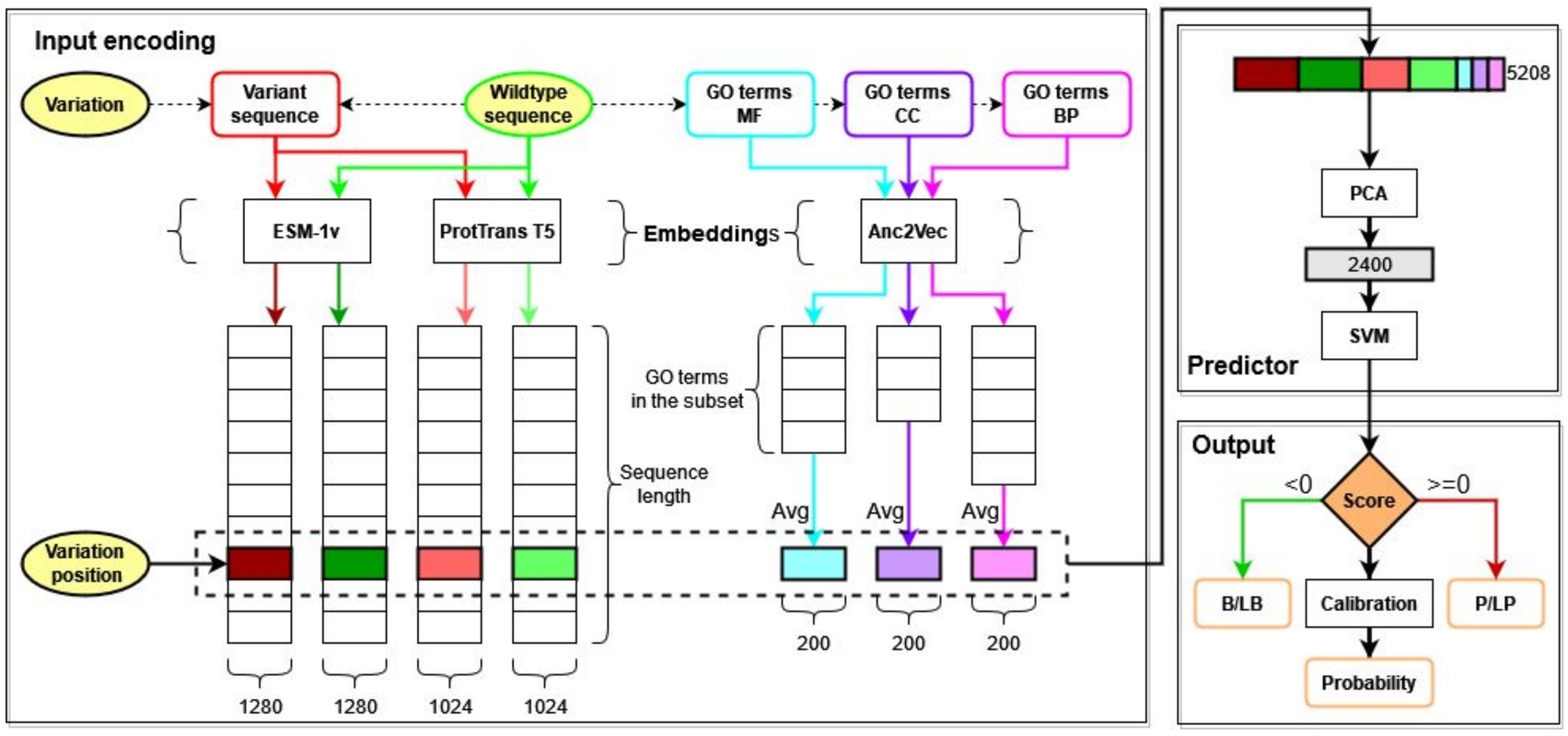
Single Nucleotide Polymorphisms (SNPs) are major drivers of human evolution. Specifically, SNPs occurring in protein-coding regions often lead to observable changes in the protein residue sequence that may have an impact at different levels, possibly setting the basis for the onset of pathologic conditions. Clear associations to diseases are still unknown for many SNPs and effective computational tools are needed to provide large-scale annotation of uncharacterized human variation data.

Here, we present E-SNPs&GO, a novel method for the prediction of variant pathogenicity from the human protein sequence that adopts recently developed embedding techniques to encode the input, in place of canonical, hand-crafted, features such as physicochemical properties or evolutionary information.

Protein Language Models

Protein Language Models aim to learn numerical vector representations of protein sequences, extracting key features that are reflected in the evolutionary conservation and in the sequence syntax. PLMs are trainable generative models aiming at reconstructing masked residues of a protein sequence starting from the context. They consist of two modules: i) the feature encoding module (encoder) provides a non-linear representation of the input and ii) the generative module (decoder) reconstructs the masked portion. Both modules are trained together on a large set, routinely in the range of hundreds of millions, of protein sequences. Depending on the problem at hand, after training, the generative part of the model (decoder) can be neglected, keeping only the encoding function that allows computing the embedding of new sequences.

Methods



Input encoding: The wild type sequence and the variant sequence are modelled with ESM-1v[1] and ProtTrans T5[2]. The Molecular Function (MF), Cellular Component (CC) and Biological Process (BP) GO functional annotations are modelled with Anc2Vec[3]. The vectors within the dashed box representing the variation position and the averaged (Avg) GO terms of the wild type sequence, are then concatenated to obtain a final representation consisting of a total of 1280*2 + 1024*2 + 200*3 = 5208 features.

Predictor: The input representation is processed with a Principal Component Analysis (PCA) to reduce the input dimensionality (from 5208 to 2400) and then a Support Vector Machine (SVM) that gives in output a single numerical score.

Output: The score is used to perform a binary classification, predicting the variation as Benign (B/LB) if the score is negative, Pathogenic (P/LP) otherwise. The score is also calibrated to obtain a pathogenicity probability that can be used to estimate the reliability of the prediction.

E-SNPs&GO webserver: https://esnpsandgo.biocomp.unibo.it/

Results

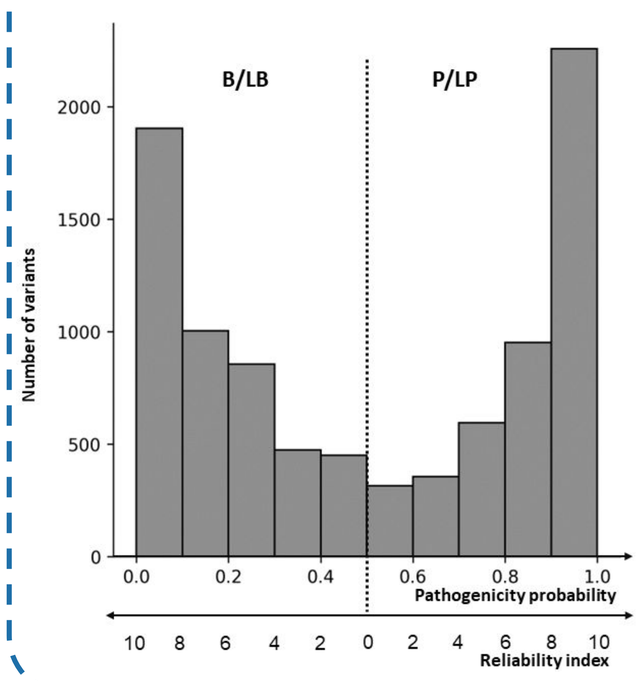
Datasets and Comparative Benchmark

Training set: 39,812 pathogenic variations and **61,334** benign variations mapped on **12,347** unique proteins. This training set was further divided into 10 balanced subsets for cross-validation.

Blind test set: 4,083 pathogenic variations and **6,183** benign variations mapped on **1,314** unique proteins. All proteins belonging to this set have less then 25% sequence identity over a pairwise alignment coverage of at least 40% with all the proteins in the training set.

Method	Q_2 (%)	Precision	Recall	F1-score	ROC-AUC	MCC
E-SNPs&GO	**86.8%**	**85.7%**	**80.1%**	**82.8%**	**85.6%**	**0.72**
SNPs&GO[4]	79.8%	84.8%	63.2%	72.4%	77.5%	0.58
MutPred2.0[5]	85.6%	78.6%	87.7%	82.9%	85.9%	0.71
PROVEAN[6]	78.2%	68.7%	83.0%	75.2%	79.0%	0.57
SIFT[7]	74.4%	62.7%	88.0%	73.2%	76.7%	0.53
PolyPhen-2[8]	72.3%	60.6%	89.5%	72.2%	75.1%	0.50

Variants of Unknown Significance



We tested E-SNPs&GO on a dataset of 2,588 proteins annotated with 9,165 Variants of Uncertain Significance (VUS) extracted from HUMSAVAR. Given that they are uncertain, we cannot assess our performances on this dataset. However, we can sample our predicted annotation in terms of probability and reliability.

3,210 out of 4,527 P/LP and 2,908 out of 4,628 B/LB predictions score with a Reliability Index greater than or equal to 6, accounting for **67%** of the VUS.

The computed probabilities are calibrated, meaning that in a set of predictions scored with probability P, we expect that the fraction of true positives is exactly P.

References

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